

Course Introduction

CSC 230 : C and Software Tools

NC State Department of Computer
Science

Teaching Staff

- Instructor: David Sturgill
 - On-campus sections and the Engineering Online section
 - Office hours in person and online via My Digital Hand (MDH)
- Usually 1 to 3 on-campus sections
- Teaching Assistants
 - Several teaching assistants and undergraduate Tas
 - Assigned to various sections
 - ... but any member of the teaching staff should be able to help with technical problems.
 - Office hours at:
<https://dbsturgi.github.io/csc230/OfficeHours.html>

Electronic Resources

- Course Homepage
 - In **Moodle**
<http://wolfware.ncsu.edu>
 - Online support for:
 - Course organization
 - Slides and code examples
 - Exercises and Exercise Submission
 - Grades
 - Review materials
- Homeworks and Project submission via **github**
- Discussion group in **Ed Discussion**
- Office hours NCSU **MyDigitalHand**

What We Will Learn

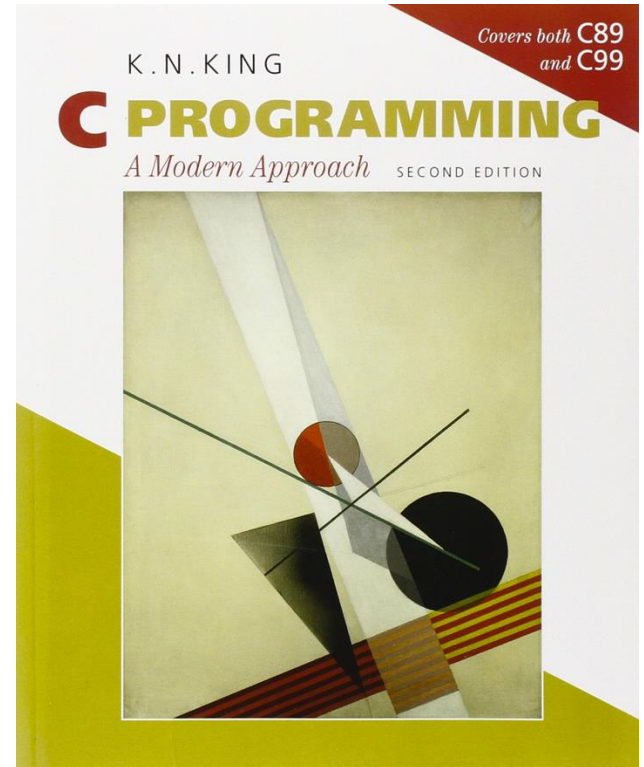
- How to design with procedures
- What's going on closer to the hardware
- Supporting library code
- Tools and techniques for building and maintaining software.

Course Requirements

- Prerequisite : CSC 216 Programming Concepts in Java
 - Programming elements
(looping, branching, types, arrays, objects)
 - Basic ADTs and Data structures
(stacks, linked lists, resizable arrays)
 - Basic algorithms
(sorting, binary search)

Required Textbook

- King, C Programming, A Modern Approach, 2nd edition
 - Covers the oldest, C89 standard
 - ... and a newer, C99 standard



Topics

- C Programming
 - Language elements
 - Types, representation and storage
 - Expressions, statements and program organization
 - Memory allocation and management
 - Tools for compilation, debugging, profiling, source control, etc.
 - Representation of program code and data
- Semester schedule with topics & readings on Moodle

Grading

- Typical NCSU Grading Scale

Category	Weight
Quizzes	8
Exercises	8
Homeworks	24
Semester Project	10
First Exam	14
Second Exam	14
Final Exam	22

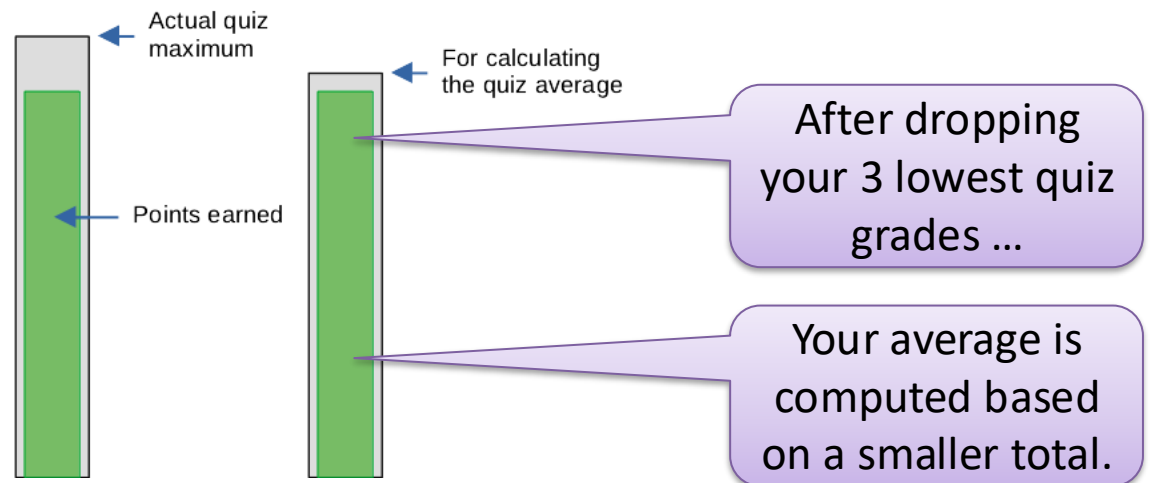
Points Earned	Letter Grade
97	A+
93	A
90	A-
87	B+
83	B
80	B-
77	C+
73	C
70	C-
67	D+
63	D
60	D-

Exams

- Preliminary exams during class
 - Exam 1, lectures 1 – 9, **at about 5 weeks**
 - Exam 2, lectures 10 – 18, **at about 11 weeks**
- Final exam: 2.5 hours, comprehensive, early during exam week
- You get to use a 3x5 inch card of notes to each exam
 - Handwritten, with your name written on it
 - Both sides if you like
 - Don't bring a card that's too large ☹️

Daily Quiz

- A quiz for almost every lecture
 - Completed online through Moodle
 - Closed book, closed notes
 - Generally due Monday at 11:59 pm
- Usually multiple choice
- Usually around 8 to 10 questions
- Drop 3 lowest quizzes ... and 5 percent worth of the remaining points



Exercises

- Small programming problems outside class
- Usually due on Monday evening, 11:59 pm
 - Typically including the last week of class
- Same as quizzes for dropping grades

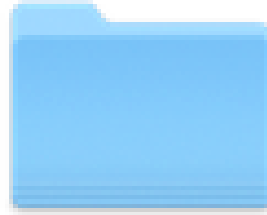
Programming Assignments

- Apply what we've learned to larger, more interesting problems
- Four Medium-Sized Homework Assignments
- Larger Semester Project
- Generally completed individually
- Submitted electronically
 - Early or on-time submission: 😊
 - Up to 24 hours late: 😐
10 percent penalty
 - Up to 48 hours late: 😐
20 percent penalty
 - After 48 hours: ☹️
No credit

Style Requirements

- We have a style guide
- Linked from the course homepage
- It says:
 - Where the curly brackets should go
 - What needs to be commented, and how
 - Policies for magic numbers, global variables, that kind of thing
- Correct style will be required on programming projects
- ... but not on exercise submissions

Your Sympathy Folder



- A catch-all for anything that can't be graded normally
 - Submission past the deadline 😞
 - Forgot to submit a file 😞
 - Submitted the wrong version of a file 😞
 - A small mistake that prevented a program from working 😞
- These items won't be graded normally
- ... but they will be considered when assigning final grades.

Special Circumstances

- Extension on a project
 - Email subject **Extension**
- Reschedule exam
 - Email subject **Exam**
- Regrade request
 - Moodle links to forms for project and exercise regrades
 - Exam regrades through GradeScope
- Sympathy folder item
 - Email subject **Sympathy**



You'll need some kind of documentation for these.

Communication

- Course Website
- Discussion and announcements via Ed Discussion
 - Between-class announcements
 - Questions and answers from students and teaching staff
 - Remember: **don't post parts of your solution**
 - Consider visiting office hours
 - Or, use a private post to the teaching staff
- Office Hours for the entire teaching staff
 - Including all sections of the class.

About Me

- Ph.D. from Cornell University
- 14 years at Baylor University
- Joined NCSU in Fall 2011
- Outside this class
 - CSC 236 (Assembly Language)
 - CSC 492 (Senior Design)
 - Programming team coach
 - Monday 5:20 – 7:10 pm
 - EB2 1203A
 - Runner
 - Trombone player
 - Unicycle non-expert
 - Daughter in grad school (computer science)
 - Another daughter freshman in engineering
 - Pet parrot



How to Enjoy C and Software Tools

- Do the exercises
 - Lots of opportunities to practice
 - Make sure you understand what we're covering
- Come to class
 - Try to understand things the first time
 - Ask questions as we go
 - Earn some easy quiz points
- Use available resources
 - Course homepage, textbook, lecture slides, examples
 - Scheduled office hours, or just stop by
 - Ed Discussion group